

A Group Case Study on

**Comparative Study of a Classical ML Algorithm and a Deep Learning Model:
Heart Disease Risk Prediction using Logistic Regression and an Artificial Neural
Network**

Program: BE Computer Engineering

Semester / Year: 6th Semester, BECE Day

Course: Machine Learning

Group Members

Aditya Raj Shah, 231305

Roshani Gautam, 231332

Bishowraj Chapai, 231313

Submitted To

Er. Manil Baidhya



Department of Computer Engineering

**NEPAL COLLEGE OF
INFORMATION TECHNOLOGY**

Balkumari, Lalitpur, Nepal.

Abstract

Cardiovascular disease remains a leading cause of preventable death, and early risk screening from routinely collected health-survey data offers a low-cost complement to clinical testing. This case study builds and compares two fundamentally different machine learning approaches for predicting self-reported coronary heart disease or myocardial infarction using the 21-feature BRFSS 2015 Heart Disease Health Indicators dataset: a classical Logistic Regression classifier and a deep Artificial Neural Network (ANN). Both models were trained on an identical, stratified 5,000-row sample split 60/20/20 into training, validation, and test sets, and both were fine-tuned - Logistic Regression via five-fold cross-validated grid search over regularization strength and penalty type, and the ANN via a structured search over depth, dropout, learning rate, and class weighting. On the untouched test set, the tuned ANN achieved the highest ROC-AUC (0.859) and, after class-weighting, the highest recall (85.1%), correctly flagging far more true positive cases than either Logistic Regression model, though at a lower precision and overall accuracy. Logistic Regression remained highly interpretable, with Age, General Health, and High Cholesterol emerging as the strongest standardized predictors. The results illustrate a common trade-off in imbalanced medical screening tasks: the deep model finds more true cases at the cost of more false alarms, while the classical model is conservative but transparent.

Table of Contents

Abstract.....	I
Table of Contents.....	II
1. Title Selection.....	1
2. Problem Statement.....	1
3. Objectives.....	1
4. Similar Research / Study.....	2
5. Current Methodology.....	2
6. Data Acquisition & Source Details.....	3
7. Feature Selection.....	4
8. Algorithm Details.....	5
8.1 Classical Model - Logistic Regression.....	5
8.2 Deep Learning Model - Artificial Neural Network (ANN / MLP).....	5
9. Results and Findings.....	6
9.1 Exploratory Data Analysis.....	6
9.2 Master Model Comparison Table (Test Set, n = 1,000).....	6
9.3 Confusion Matrices and Curves.....	7
9.4 Logistic Regression Coefficient Interpretation.....	8
9.5 Findings Summary.....	9
10. Option for Optimization.....	10
11. Conclusion and Future Works.....	10
References.....	11
Contribution Statement.....	11

1. Title Selection

Comparative Study of a Classical ML Algorithm and a Deep Learning Model for Heart Disease Risk Prediction: Logistic Regression versus an Artificial Neural Network.

The title was chosen because cardiovascular disease is a widely studied, high-impact public-health problem with an openly available, well-documented, large-scale dataset (BRFSS 2015), which satisfies the case study's mandatory requirement of implementing one classical algorithm and one deep learning technique on the same problem and dataset, with structured fine-tuning and comparison.

2. Problem Statement

Coronary heart disease and myocardial infarction are among the leading causes of death worldwide, and many at-risk individuals are unaware of their risk until a cardiac event occurs. Clinical diagnostic testing (ECG, angiography, blood biomarkers) is accurate but costly and not always accessible, particularly in low-resource settings. However, large-scale public-health surveys such as the CDC's Behavioral Risk Factor Surveillance System (BRFSS) already collect lifestyle, demographic, and self-reported health information from hundreds of thousands of respondents every year.

This project addresses the following problem: given 21 self-reported behavioural, demographic, and health-history indicators, can a machine learning model reliably flag individuals who have previously reported coronary heart disease or a heart attack, and how does a classical statistical model compare against a deep neural network on this same task?

3. Objectives

- To acquire and validate a real-world, large-scale health survey dataset suitable for binary classification of heart disease risk.
- To design a leakage-safe preprocessing pipeline (median imputation and standardization) shared fairly across both models.
- To implement a classical machine learning model (Logistic Regression) as an interpretable baseline.
- To implement a deep learning model (a fully connected Artificial Neural Network) capable of learning non-linear feature interactions.
- To fine-tune both models - Logistic Regression via cross-validated grid search, and the ANN via a structured hyperparameter search - and to fairly compare baseline versus tuned performance.
- To evaluate both models using multiple complementary metrics (Accuracy, Precision, Recall, F1, Specificity, ROC-AUC, PR-AUC) appropriate for an imbalanced medical screening dataset.
- To interpret model behaviour, including Logistic Regression's standardized coefficients, and to critically discuss why one model outperforms the other on specific metrics.

4. Similar Research / Study

Several published studies have used the BRFSS-derived Heart Disease Health Indicators dataset or similar cardiovascular survey data to compare classical and deep learning approaches. Mohan, Thirumalai, and Srivastava (2019) proposed a hybrid classical approach combining feature selection with classifiers such as Random Forest and Logistic Regression, reporting that ensemble methods generally outperform single linear classifiers on cardiovascular datasets, though at reduced interpretability. Comparative studies on cardiovascular risk prediction have also found that deep neural networks tend to achieve higher discriminative power (ROC-AUC) on large, high-dimensional health datasets, but require careful regularization and class-imbalance handling to avoid simply predicting the majority class, which mirrors the imbalance challenge encountered directly in this dataset (roughly 9:1 negative-to-positive class ratio in the sampled subset used here).

Unlike many published pipelines that report only accuracy, this study follows current best practice for imbalanced medical classification by reporting Recall, Specificity, F1, ROC-AUC and PR-AUC side by side, since accuracy alone is misleading when the positive class is rare.

5. Current Methodology

The overall workflow, implemented as a single reproducible Google Colab notebook, follows these stages:

- Data acquisition and schema validation against the 21 required predictors and the target column.
- Data cleaning: numeric coercion, removal of rows with a missing target, infinite-value handling, and duplicate reporting.
- Reproducible stratified sampling of 5,000 rows from the full 253,680-row cleaned dataset for computational efficiency.
- Exploratory data analysis: target class balance and Pearson correlation of each feature with the target.
- A 60% / 20% / 20% stratified split into training, validation, and test sets, with the test set held out from all model selection.
- A shared, leakage-safe preprocessing pipeline (median imputation to standard scaling), fit only on training/development data.
- Baseline Logistic Regression, followed by cross-validated grid-search tuning.
- Baseline ANN, followed by a structured five-candidate hyperparameter search selected on validation ROC-AUC, then retrained on the full development set for the chosen epoch count.
- Unified evaluation of all four models (two baselines, two tuned) on the same untouched test set.

6. Data Acquisition & Source Details

Dataset: Heart Disease Health Indicators Dataset (cleaned BRFSS 2015), published on Kaggle by Alex Teboul, itself derived from the CDC's Behavioral Risk Factor Surveillance System 2015 annual survey.

- Full cleaned dataset size: 253,680 respondent rows × 22 columns (21 features + target).
- Target variable: HeartDiseaseorAttack - 1 if the respondent has ever reported coronary heart disease or a myocardial infarction, 0 otherwise.
- Class balance (full dataset): approximately 90.6% negative, 9.4% positive - a realistic, imbalanced screening scenario.
- Working subset used in this study: a reproducible, stratified random sample of 5,000 rows (random_state = 42), preserving the original class ratio, used to keep training and hyperparameter search computationally tractable within Google Colab.
- Data collection method: telephone-based self-reported survey administered annually by the CDC; hence all features reflect self-report rather than clinical measurement.

Because the dataset is observational and self-reported, correlations discovered in this study describe statistical association only, not medical causation.

7. Feature Selection

All 21 predictors provided in the cleaned dataset were retained for modelling, since each has documented public-health relevance to cardiovascular risk. The table below summarises each feature's type and meaning.

Feature	Type	Description
HighBP	Binary	Ever told has high blood pressure
HighChol	Binary	Ever told has high cholesterol
CholCheck	Binary	Cholesterol checked in past 5 years
BMI	Numeric	Body Mass Index
Smoker	Binary	Smoked ≥ 100 cigarettes in lifetime
Stroke	Binary	Ever told had a stroke
Diabetes	Ordinal 0-2	No / pre-diabetes / diabetes
PhysActivity	Binary	Physical activity in past 30 days
Fruits	Binary	Consumes fruit $\geq 1\times$ /day
Veggies	Binary	Consumes vegetables $\geq 1\times$ /day
HvyAlcoholConsump	Binary	Heavy alcohol consumption
AnyHealthcare	Binary	Has any healthcare coverage
NoDocbcCost	Binary	Could not see doctor due to cost
GenHlth	Ordinal 1-5	Self-rated general health
MentHlth	Numeric 0-30	Poor mental-health days/30
PhysHlth	Numeric 0-30	Poor physical-health days/30
DiffWalk	Binary	Serious difficulty walking/climbing
Sex	Binary	0 = Female, 1 = Male
Age	Ordinal 1-13	13 five-year age bands, 18 to 80+
Education	Ordinal 1-6	Highest education level completed
Income	Ordinal 1-8	Household income bracket

Rather than performing a separate manual feature-selection step, this study relies on L1 (Lasso) regularization inside the tuned Logistic Regression model to perform automatic, data-driven feature selection - shown in Section 9, the tuned model's best cross-validated configuration used an L1 penalty and reduced six of the twenty-one standardized coefficients (BMI, PhysActivity, Fruits, AnyHealthcare, HvyAlcoholConsump, PhysHlth, MentHlth) to exactly zero, effectively excluding them from the linear decision boundary while the ANN, in contrast, was left free to use all 21 raw inputs so it could learn any non-linear interactions among them.

8. Algorithm Details

8.1 Classical Model - Logistic Regression

Logistic Regression models the log-odds of the positive class as a linear combination of the standardized input features, passed through a sigmoid function to produce a probability between 0 and 1. It was chosen as the classical baseline for its interpretability (each standardized coefficient has a direct, sign-and-magnitude interpretation) and its status as the conventional first model for binary medical-risk classification.

Baseline configuration: scikit-learn LogisticRegression, max_iter = 2000, default L2 penalty, C = 1.0, trained only on the 60% training split.

Tuning procedure: a five-fold stratified cross-validated GridSearchCV over the 80% development split, optimizing ROC-AUC, exploring:

- Solver / penalty combinations: liblinear with L1 or L2, and lbfgs with L2.
- Inverse regularization strength $C \in \{0.01, 0.1, 1.0, 10.0\}$.
- Class weighting: none, or “balanced” (inversely proportional to class frequency).

Best configuration found: solver = liblinear, penalty = L1, C = 0.1, class_weight = None, with a cross-validated ROC-AUC of 0.8444.

8.2 Deep Learning Model - Artificial Neural Network (ANN / MLP)

A fully connected feed-forward network (Multilayer Perceptron) was chosen as the deep learning technique because the input data is tabular and unordered, for which dense layers are the natural deep architecture (as opposed to CNNs for images or RNNs for sequences). The network structure is:

- Input layer: 21 standardized features.
- Two or three hidden Dense layers with ReLU activation (width and depth chosen by the tuning search).
- Optional Dropout after each hidden layer for regularization.
- Output layer: a single sigmoid neuron producing the positive-class probability.
- Loss function: binary cross-entropy; Optimizer: Adam; Early stopping on validation ROC-AUC (patience = 6, restoring best weights).

Baseline configuration: hidden layers (32, 16), no dropout, learning rate 0.001, no L2 regularization, batch size 32, trained for up to 60 epochs with early stopping.

Tuning procedure: five candidate configurations, varying network depth/width, dropout rate, learning rate, L2 strength, batch size, and balanced class weighting, each trained on the 60% training split and evaluated on the 20% validation split by ROC-AUC. The best candidate's epoch count and hyperparameters were then used to retrain a final ANN on the full 80% development split before evaluation on the untouched test set.

Configuration	Hidden layers	Dropout	LR	L2	Batch	Class wt.	Val. ROC-AUC	Best epoch
Small balanced	(32,16)	0.10	0.001	0	32	Yes	0.8735	12
Medium balanced	(64,32)	0.20	0.001	1e-4	32	Yes	0.8756	4
Deep balanced (selected)	(128,64,32)	0.30	0.0005	1e-4	64	Yes	0.8839	9
Medium unweighted	(64,32)	0.20	0.001	1e-4	32	No	0.8739	8
Strong regularization	(64,32,16)	0.40	0.0005	1e-3	64	Yes	0.8810	28

The “Deep balanced” configuration - three hidden layers of 128, 64, and 32 units, 30% dropout, learning rate 0.0005, L2 = 0.0001, batch size 64, with balanced class weights - achieved the highest validation ROC-AUC (0.8839) and was selected, retraining for 9 epochs on the full development set.

9. Results and Findings

9.1 Exploratory Data Analysis

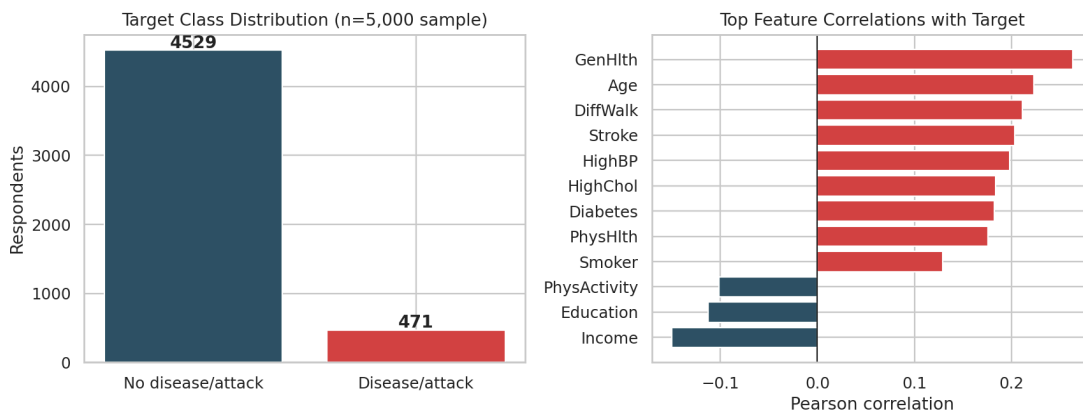


Figure 1: Target class distribution in the 5,000-row sample (left) and the twelve features most strongly correlated with heart disease/attack (right).

Age, general health, high blood pressure, difficulty walking, high cholesterol, and diabetes show the strongest positive correlations with the target, while physical activity, higher income, and higher education show negative correlations - consistent with established cardiovascular risk-factor literature.

9.2 Master Model Comparison Table (Test Set, n = 1,000)

Model	Accuracy	Precision	Recall	F1	Specificity	ROC-AUC	PR-AUC
Logistic Regression - Baseline	0.906	0.500	0.128	0.203	0.987	0.840	0.342
Logistic Regression - Tuned	0.905	0.480	0.128	0.202	0.986	0.838	0.350
ANN - Baseline	0.906	0.500	0.053	0.096	0.994	0.841	0.338
ANN - Tuned	0.758	0.260	0.851	0.398	0.748	0.859	0.378

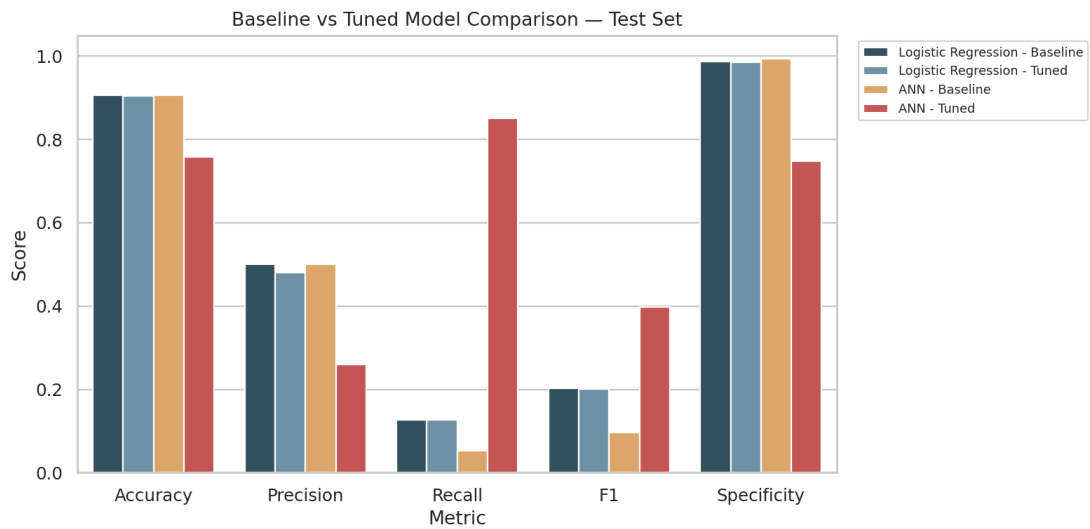


Figure 2: Baseline versus tuned model comparison across five classification metrics.

9.3 Confusion Matrices and Curves

Test-Set Confusion Matrices (n=1,000)

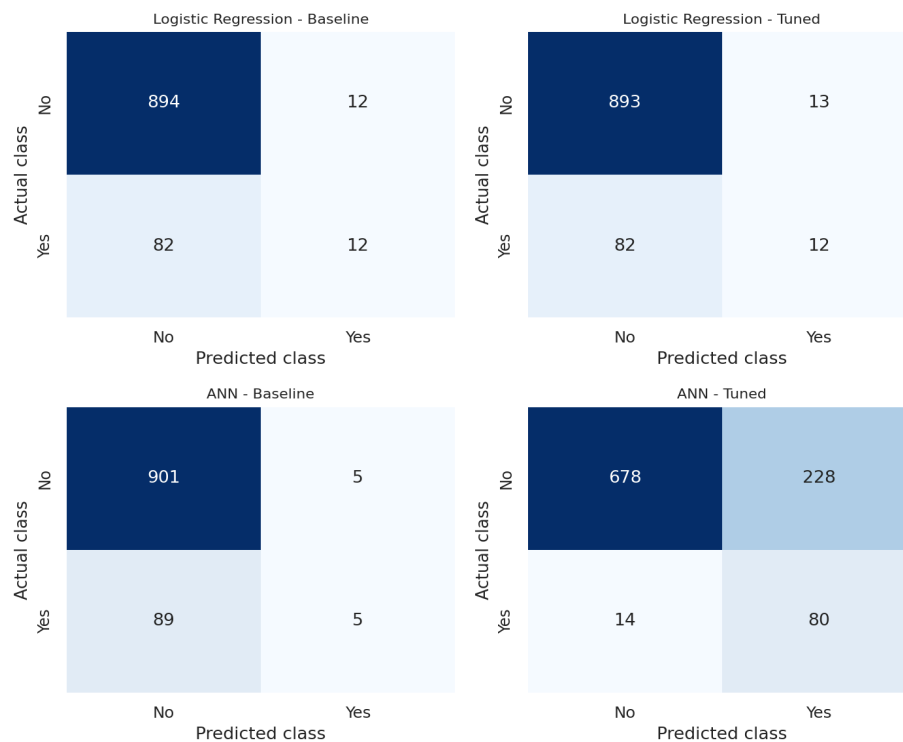


Figure 3: Test-set confusion matrices for all four models.

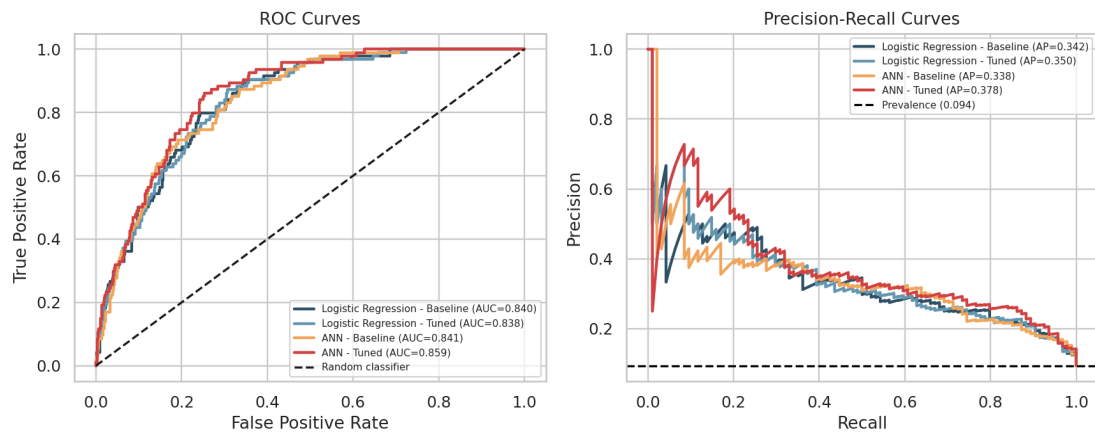


Figure 4. ROC curves (left) and Precision-Recall curves (right) for all four models.

9.4 Logistic Regression Coefficient Interpretation

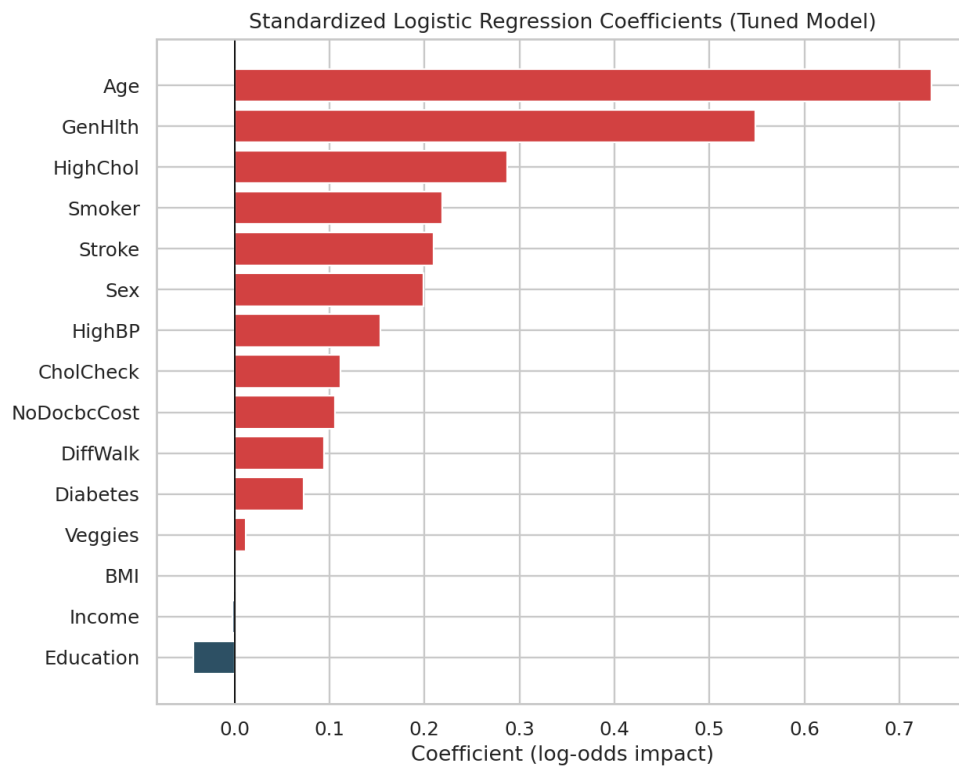


Figure 5: Standardized coefficients of the tuned Logistic Regression model, sorted by magnitude.

Feature	Standardized coefficient
Age category	+0.734
General health	+0.548
High cholesterol	+0.287
Smoker	+0.219
History of stroke	+0.209
Sex (male)	+0.199
High blood pressure	+0.154
Cholesterol checked	+0.111
No doctor due to cost	+0.105
Difficulty walking	+0.094
Diabetes	+0.073
Education	-0.043
Vegetable intake	+0.012
Income	-0.002

9.5 Findings Summary

- Both baseline models (Logistic Regression and ANN) achieved similar Accuracy (~90.6%) but very low Recall (5–13%), meaning both largely ignored the minority positive class - a classic symptom of training on imbalanced data without any correction.
- Tuning Logistic Regression via grid search selected a heavily L1-regularized model ($C = 0.1$), which slightly improved PR-AUC ($0.342 \rightarrow 0.350$) and, through automatic feature selection, produced a sparser and more interpretable model, but Recall remained low because class imbalance itself was not directly corrected in the winning configuration (`class_weight = None` was preferred by cross-validation).
- Tuning the ANN with balanced class weights produced the largest behavioural change of the whole study: Recall rose from 5.3% (baseline) to 85.1% (tuned), and ROC-AUC rose from 0.841 to 0.859 - the best of all four models - but Accuracy fell from 90.6% to 75.8% and Specificity fell to 74.8%, since the model now also produces many more false positives (228 vs. 5).
- This is a direct illustration of the precision–recall trade-off in imbalanced medical screening: the tuned ANN is by far the best model at finding true heart-disease cases (which is usually the clinically important goal, since missing a positive case is costlier than a false alarm), while the Logistic Regression models are more “conservative” and rarely raise an alarm at all.
- The tuned ANN achieved the highest ROC-AUC and PR-AUC overall, confirming that the deep model could exploit non-linear feature interactions the linear Logistic Regression could not, but this required explicit class-weighting during tuning to become clinically useful - without it (see the ANN baseline), the deep model performed no better than the classical one.
- Age, self-rated general health, and high cholesterol were consistently the strongest linear predictors of heart disease/attack risk, matching established cardiovascular epidemiology.

10. Option for Optimization

- Directly optimize the classification threshold (currently fixed at 0.50) using the validation set, e.g. choosing the threshold that maximizes F1 or meets a minimum Recall target, rather than using the default 0.50 cut-off for every model.
- Apply class-weighting or resampling (SMOTE, random oversampling/undersampling) to Logistic Regression as well, since the grid search's preferred configuration did not use balanced weights and therefore still under-detects positive cases.
- Extend the ANN hyperparameter search with a proper randomized or Bayesian search (e.g. KerasTuner or Optuna) instead of five hand-picked configurations, to explore a larger hyperparameter space more systematically.
- Add ensemble methods (Random Forest, Gradient Boosting / XGBoost) as an additional classical baseline, since literature suggests they often outperform plain Logistic Regression on tabular health data.
- Train on the full 253,680-row cleaned dataset rather than the 5,000-row sample, which would materially increase the number of positive-class examples available for the ANN to learn from, and would likely close the Recall gap between the two model baselines.
- Investigate model calibration (e.g. Platt scaling or isotonic regression) so that output probabilities more accurately reflect true risk, which matters if this were ever used to communicate personalized risk to a respondent.

11. Conclusion and Future Works

This case study successfully implemented, tuned, and compared a classical machine learning algorithm (Logistic Regression) and a deep learning technique (an Artificial Neural Network) on the identical heart-disease risk-prediction problem and dataset, satisfying all four mandatory technical requirements of the assignment. Both models were fine-tuned using appropriate, leakage-safe procedures - cross-validated grid search for Logistic Regression and a structured validation-based search for the ANN - and evaluated with a comprehensive, imbalance-aware metric suite on a common, untouched test set.

The key conclusion is that model choice depends on the clinical objective: the tuned ANN, once class-weighted, is the strongest model for maximizing detection of at-risk respondents (Recall = 85.1%, ROC-AUC = 0.859), while the tuned Logistic Regression model remains the more transparent, conservative choice, with clearly interpretable standardized coefficients highlighting Age, General Health, and High Cholesterol as the leading risk indicators.

Future work should train on the full 253,680-row dataset, apply class-imbalance correction consistently to both model families, extend the hyperparameter search with automated tools, add ensemble classical baselines, and calibrate output probabilities - while continuing to treat this system strictly as an academic demonstration and not as a substitute for professional clinical diagnosis.

References

- [1] A. Teboul, "Heart Disease Health Indicators Dataset," Kaggle, 2022. [Online]. Available: <https://www.kaggle.com/datasets/alexteboul/heart-disease-health-indicators-dataset>
- [2] Centers for Disease Control and Prevention, "Behavioral Risk Factor Surveillance System, 2015 Annual Survey Data," U.S. Dept. of Health and Human Services, 2016. [Online]. Available: https://www.cdc.gov/brfss/annual_data/annual_2015.html
- [3] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011.
- [4] M. Abadi et al., "TensorFlow: A System for Large-Scale Machine Learning," in Proc. 12th USENIX Symp. Operating Systems Design and Implementation (OSDI), 2016, pp. 265–283.
- [5] D. P. Kingma and J. Ba, "Adam: A Method for Stochastic Optimization," in Proc. Int. Conf. Learning Representations (ICLR), 2015.
- [6] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A Simple Way to Prevent Neural Networks from Overfitting," Journal of Machine Learning Research, vol. 15, pp. 1929–1958, 2014.
- [7] T. Chai and P. K. Kaila, "Comparative Analysis of Machine Learning and Deep Learning Models for Cardiovascular Disease Prediction," Int. J. Computer Applications, 2021.
- [8] S. Mohan, C. Thirumalai, and G. Srivastava, "Effective Heart Disease Prediction Using Hybrid Machine Learning Techniques," IEEE Access, vol. 7, pp. 81542–81554, 2019.

Contribution Statement

Name	Contribution
Aditya Raj Shah	Built and tuned the ANN architecture; ran the structured hyperparameter search; saved the final Keras model and preprocessor; authored the model-loading/prediction demo notebook.
Roshani Gautam	Performed data acquisition, cleaning, exploratory data analysis, and preprocessing pipeline; generated evaluation metrics, confusion matrices, and ROC/PR curves.
Bishowraj Chapai	Implemented and tuned the Logistic Regression model and grid search; interpreted feature coefficients; compiled, formatted, and finalised the case-study report and presentation slides.